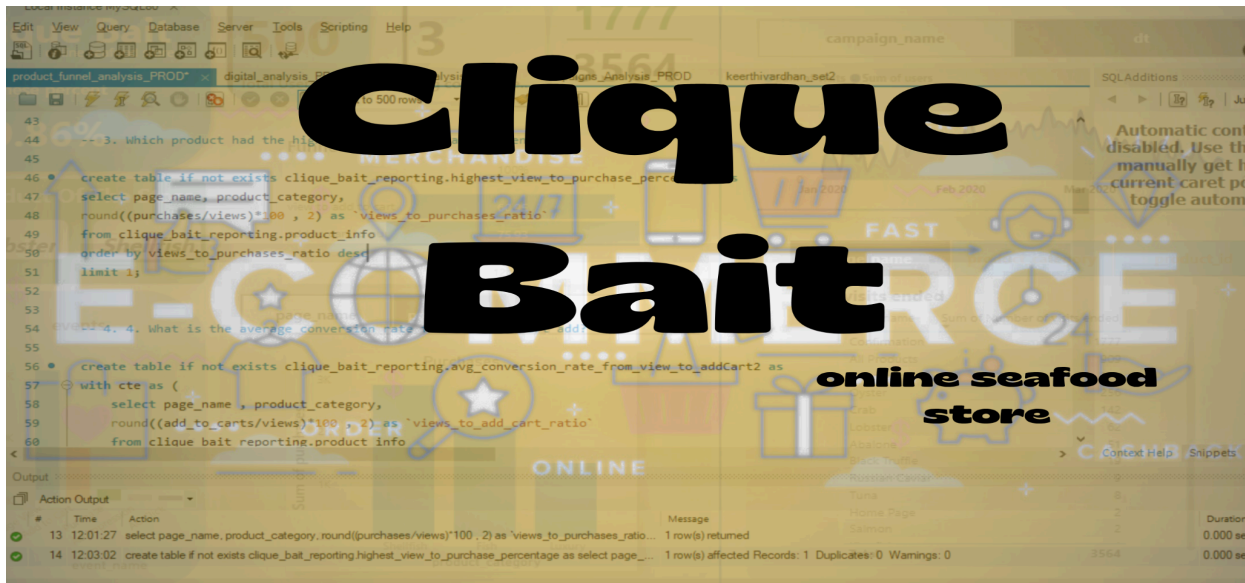




Introduction

Clique Bait is not like your regular online seafood store - the founder and CEO Keerthi, was also a part of a digital data analytics team and wanted to expand his knowledge into the seafood industry!

*In this case study - you are required to support Keerthi's vision and analyse his dataset and come up with creative solutions to **calculate funnel fallout rates** for the Clique Bait online store.*

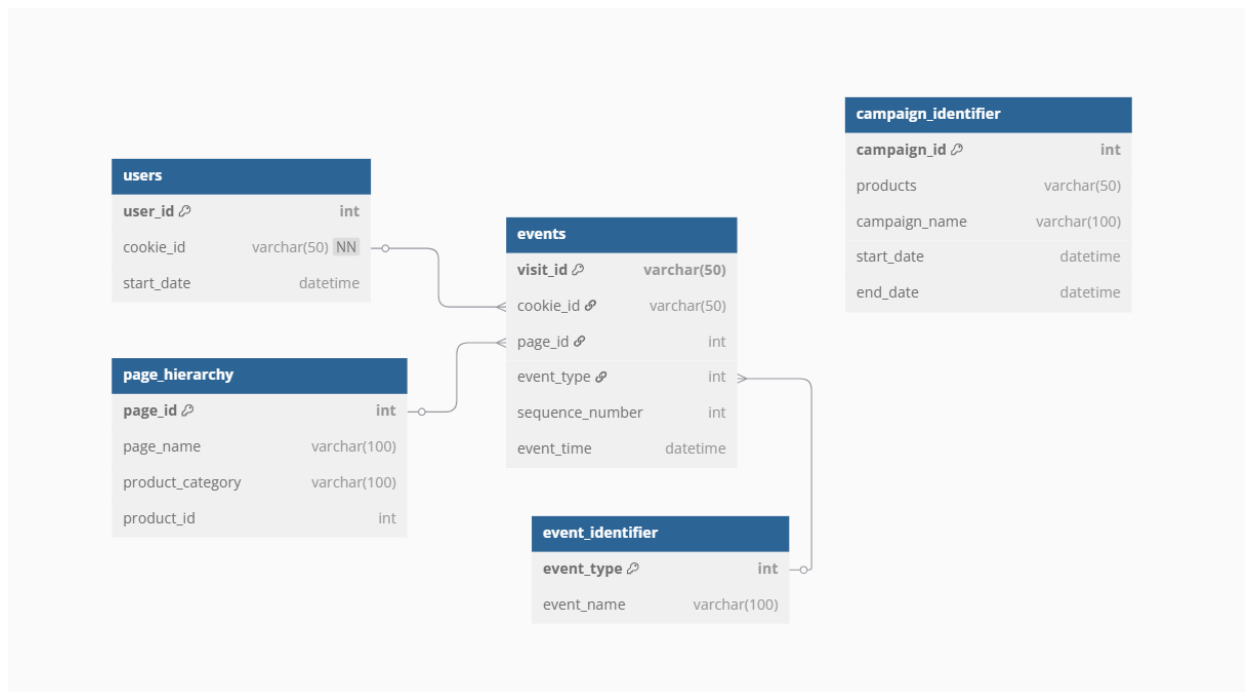
Mentor : Nusrath Syed

Name : Keerthivardhan Tekulapelli

Date : 11-05-2025

Schema : ER Diagram

1. Enterprise Relationship Diagram



2. Digital Analysis

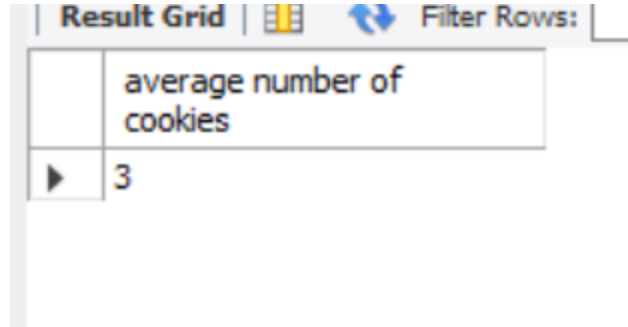
1. How many users are there?

- To find total number of users we need to count select distinct users from **users** table
- [Code](#)

```
select count(distinct user_id) as `Total number of users` from
clique_bait.users;
```

2. How many cookies does each user have on average?

- To find average cookies each user has, we need to first find the total number of cookies each user has and then find avg



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains one row with the column header 'average number of cookies' and a value of 3. A small play button icon is visible in the first column of the row.

	average number of cookies
▶	3

- [Code](#)

```
create table if not exists clique_bait_reporting.avg_cookies_per_user
as
with cte as (
    select user_id , count(cookie_id) as cnt
    from clique_bait.users
    group by user_id
)
select floor(avg(cnt)) as `average number of cookies` from cte;
```

3. What is the unique number of visits by all users per month?

- events table consists all the logs generated for each visit there will be unique visit_id that gets generated so, our job is to count how many unique visit_ids are generated for each month

Result Grid			Filter Rows:
	month_number	number_of_visits	
▶	1	876	
	2	1488	
	3	916	
	4	248	
	5	36	

-
- [Code](#)

```
select
    month(event_time) as month_number,
    count(distinct visit_id) as number_of_visits
from clique_bait.events
group by month_number;
```

4. What is the number of events for each event type?

- event table consists of all the logs/events generated in website throughout its life line join event and event_identifier to know type of event and we group on event_type/event_name aggregate the count() to know number of events

Result Grid			Filter Rows:
	event_name	number_of_events	
▶	Page View	20928	
	Add to Cart	8451	
	Purchase	1777	
	Ad Impression	876	
	Ad Click	702	

-
- [Code](#)

```
select
```

```
ed.event_name,
count(*) as `number_of_events`
from clique_bait.events e, clique_bait.event_identifier ed
where e.event_type = ed.event_type
group by event_name;
```

-

5. What is the percentage of visits which have a purchase event?

- events table contains only event type but to understand what is that event we need to join with event_identifier table
- To solve this problem
 - we should know whether purchase is done in a visit or not
 - So, we need to know whether, purchase event is generated in visit or not
- for this, we will group on each visit, then we can find whether, it contains purchase event or not

	purchase_event_percent
▶	49.86

-

- [Code](#)

```
with cte as (
  select visit_id,
         max(case when event_name = 'Purchase' then 1 else 0 end)
  as `purchased_or_not`
  from
    clique_bait.events e, clique_bait.event_identifier ed
  where e.event_type = ed.event_type
  group by e.visit_id
```

```

)
select
  round((sum(purchased_or_not=1)/count(visit_id))*100 , 2) as
`purchase_event_percent`
from cte;

```

6. What is the percentage of visits which view the checkout page but do not have a purchase event?

Result Grid		Filter Rows:
	checkout but not purchased %	
▶	15.50	

- [Code](#)

```

with cte as (
  select
    visit_id,
    max(case when ed.event_name = 'Purchase' then 1 else 0 end) as `purchased`,
    max(case when p.page_name = 'Checkout' then 1 else 0 end) as `visited_checkout`
  from
    clique_bait.events e, clique_bait.event_identifier ed, clique_bait.page_hierarchy p
  where
    e.event_type = ed.event_type and
    e.page_id = p.page_id
  group by e.visit_id
)
select
  round(
    (sum(visited_checkout = 1 & purchased = 0 )/

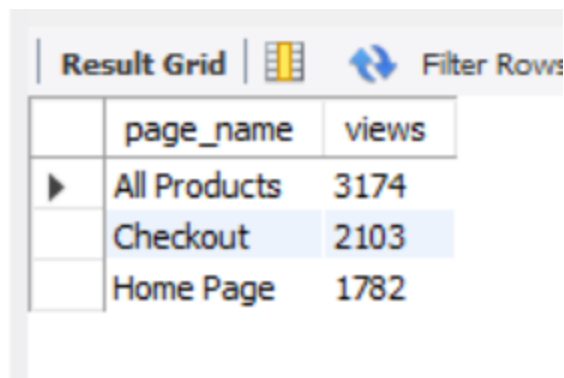
```

```
sum(visited_checkout))*100
, 2) as `checkout but not purchased %`
from cte;
```

7. What are the top 3 pages by number of views?

- Counts how often each page was viewed, selects the top 3, and joins with the page_hierarchy table to display their names.

-



	page_name	views
▶	All Products	3174
	Checkout	2103
	Home Page	1782

-

- [Code](#)

```
select
page_name,
count(distinct visit_id) as `views`
from clique_bait.events e join clique_bait.page_hierarchy p
on e.page_id = p.page_id
group by page_name
order by views desc
limit 3;
```

8. What is the number of views and cart adds for each product category?

- Aggregates the total number of “Page View” and “Add to Cart” events grouped by product category.

-

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Conte	
	product_category	number_of_views	number_of_add_to_carts
	Luxury	3032	1870
	Shellfish	6204	3792
	Fish	4633	2789

-
- [Code](#)

```
select
  p.product_category,
  sum(case when event_name = 'Page View' then 1 else 0 end) as
`number_of_views`,
  sum(event_name = 'Add to Cart') as `number_of_add_to_carts`
from
  clique_bait.events e , clique_bait.page_hierarchy p,
  clique_bait.event_identifier ed
where
  e.page_id = p.page_id and
  e.event_type = ed.event_type and
  product_category is not null
group by product_category;
```

9. What are the top 3 products by purchases?

- Identifies add-to-cart events that later led to a purchase, then ranks products by the number of successful purchases and returns the top 3.

	product_id	product_category	page_name	purchases
	7	Shellfish	Lobster	754
	9	Shellfish	Oyster	726
	8	Shellfish	Crab	719

-
- [Code](#)

```
with cte as (
  select
```



```
        e.*,
        ed.event_name,
        p.product_id,
        p.page_name,
        p.product_category,
        max(
            case
                when event_name = 'Purchase'
                then 1
                else 0
            end
        )
        over (partition by visit_id ) as
`purchased`

from
    clique_bait.events e,
    clique_bait.event_identifier ed,
    clique_bait.page_hierarchy p
where
    e.event_type= ed.event_type and
    e.page_id = p.page_id
)
select
product_id ,
product_category,
page_name,
count(*) as `purchases`
from cte
where
    event_name = 'Add to Cart' and
    purchased = 1
group by product_id,product_category,page_name
order by purchases desc
limit 3;
```

3. Product Funnel Analysis

- For this part, i have stored this output as **product_info** table
 - It consists of number of cart_adds, purchases and abandoned of each and every product
- [Code](#)

```
with cte as (  
  select  
    e.*,  
    ed.event_name,  
    p.page_name,  
    p.product_category,  
    p.product_id  
  from events e, event_identifier ed, page_hierarchy p  
  where e.event_type = ed.event_type and e.page_id = p.page_id  
)  
cte2 as (  
  select page_name,  
    product_id,  
    product_category,  
    case when event_name = 'Add to Cart' then visit_id end as  
  `cart_id`,  
    case when event_name = 'Page View' then visit_id end as `viewed`  
  from cte  
  where product_id is not null  
)  
cte3 as (  
  select visit_id as `purchased_visit_id`  
  from cte  
  where event_name = 'Purchase'  
)  
select  
  cte2.page_name,  
    cte2.product_id,  
    product_category,  
  COUNT(DISTINCT cte2.viewed) AS `views`,
```

```

COUNT(DISTINCT cte2.cart_id) AS `add_to_carts`,
COUNT(distinct cte3.purchased_visit_id) AS `purchases`,
count(DISTINCT cte2.cart_id) - COUNT(distinct
cte3.purchased_visit_id) as `Abandoned`
from
cte2
left join cte3 on cte2.cart_id = cte3.purchased_visit_id
group by
cte2.page_name,
    product_id,
    product_category;

```

1. How many times was each product added to cart?

- Retrieves total “Add to Cart” counts for each product from a product_info table.

	page_name	product_id	product_category	add_to_carts
▶	Abalone	6	Shellfish	932
	Black Truffle	5	Luxury	924
	Crab	8	Shellfish	949
	Kingfish	2	Fish	920
	Lobster	7	Shellfish	968
	Oyster	9	Shellfish	943
	Russian Caviar	4	Luxury	946
	Salmon	1	Fish	938
	Tuna	3	Fish	931

-
- [Code](#)

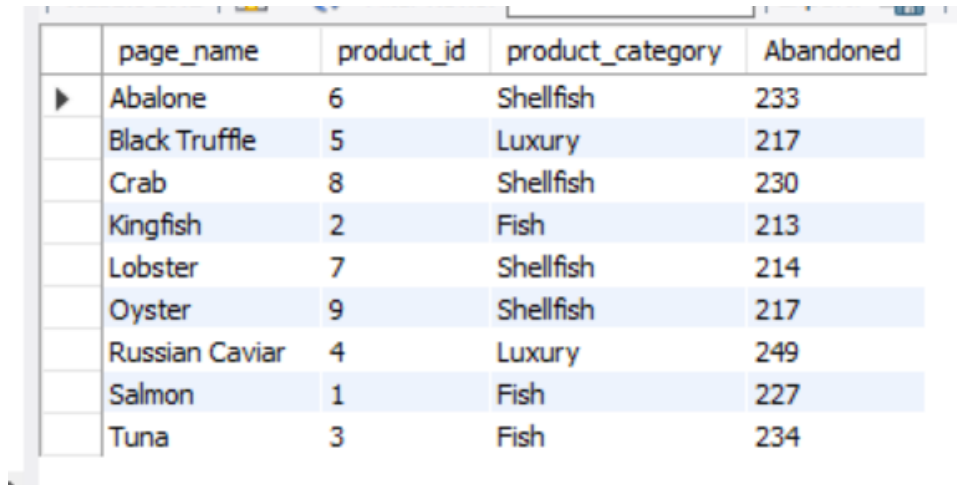
```

select
    page_name , product_id , product_category , add_to_carts
from clique_bait_reporting.product_info;

```

2. How many times was each product added to a cart but not purchased (abandoned)?

- Shows, per product, the count of cart adds that did not lead to a purchase using precomputed "Abandoned" values.



	page_name	product_id	product_category	Abandoned
▶	Abalone	6	Shellfish	233
	Black Truffle	5	Luxury	217
	Crab	8	Shellfish	230
	Kingfish	2	Fish	213
	Lobster	7	Shellfish	214
	Oyster	9	Shellfish	217
	Russian Caviar	4	Luxury	249
	Salmon	1	Fish	227
	Tuna	3	Fish	234

- [Code](#)

```
select
    page_name , product_id , product_category , Abandoned
from clique_bait_reporting.product_info;
```

3. How many times was each product purchased?

- Displays the number of successful purchases for each product from the reporting table.

Result Grid				
		Filter Rows:		Export:  Wr
	page_name	product_id	product_category	purchases
▶	Abalone	6	Shellfish	699
	Black Truffle	5	Luxury	707
	Crab	8	Shellfish	719
	Kingfish	2	Fish	707
	Lobster	7	Shellfish	754
	Oyster	9	Shellfish	726
	Russian Caviar	4	Luxury	697
	Salmon	1	Fish	711
	Tuna	3	Fish	697

-
- [Code](#)

```
select
    page_name , product_id , product_category , purchases
from clique_bait_reporting.product_info;
```

Result Grid							
		Filter Rows:		Export: 	Wrap Cell Content: 		
	page_name	product_id	product_category	views	add_to_carts	purchases	Abandoned
▶	Abalone	6	Shellfish	1525	932	699	233
	Black Truffle	5	Luxury	1469	924	707	217
	Crab	8	Shellfish	1564	949	719	230
	Kingfish	2	Fish	1559	920	707	213
	Lobster	7	Shellfish	1547	968	754	214
	Oyster	9	Shellfish	1568	943	726	217
	Russian Caviar	4	Luxury	1563	946	697	249
	Salmon	1	Fish	1559	938	711	227
	Tuna	3	Fish	1515	931	697	234

4. Which product had the most views, cart adds and purchases?

- Returns the product that ranks highest across all three metrics: views, add-to-carts, and purchases.

	page_name	product_id	product_category	views	add_to_carts	purchases	Abandoned
▶	Oyster	9	Shellfish	1568	943	726	217

-
- [Code](#)

```
select * from clique_bait_reporting.product_info
order by views desc, add_to_carts desc , purchases desc
```

```
limit 1;
```

5. Which product was most likely to be abandoned?

- Identifies the product with the highest number of cart additions that were not followed by a purchase.

	page_name	product_id	product_category	views	add_to_carts	purchases	Abandoned
▶	Russian Caviar	4	Luxury	1563	946	697	249

- [Code](#)

```
select * from clique_bait_reporting.product_info
order by Abandoned desc
limit 1;
```

6. Which product had the highest view to purchase percentage?

- Calculates and ranks products by their conversion rate from views to purchases, showing the highest.

	page_name	product_category	views_to_purchases_ratio
▶	Lobster	Shellfish	48.74

- [Code](#)

```
select page_name, product_category,
round((purchases/views)*100 , 2) as `views_to_purchases_ratio`
from clique_bait_reporting.product_info
order by views_to_purchases_ratio desc
limit 1;
```

7. What is the average conversion rate from view to cart add?

- Computes the average percentage of views that resulted in cart additions across all products.

	average_conversion_rate
▶	60.950000

- [Code](#)

```
with cte as (
  select page_name , product_category,
  round((add_to_carts/views)*100 , 2) as `views_to_add_cart_ratio`
  from clique_bait_reporting.product_info
)
```

```
select avg(views_to_add_cart_ratio) as `average_conversion_rate` from
cte;
```

8. What is the average conversion rate from cart add to purchase?

- Calculates the average percentage of cart additions that ended in a purchase across all products.

	average_conversion_rate
▶	75.928889

- [Code](#)

```
with cte as (
  select page_name , product_category,
  round((purchases/add_to_carts)*100 , 2) as
`add_cart_ratio_to_purchases`
  from clique_bait_reporting.product_info
)
select avg(add_cart_ratio_to_purchases) as `average_conversion_rate`
from cte;
```

4. Campaigns Analysis

Generate a table that has 1 single row for every unique visit_id record and has the following

columns:

- user_id
- visit_id
- visit_start_time: the earliest event_time for each visit
- page_views: count of page views for each visit
- cart_adds: count of product cart add events for each visit

- purchase: 1/0 flag if a purchase event exists for each visit
- campaign_name: map the visit to a campaign if the visit_start_time falls between the start_date and end_date
- impression: count of ad impressions for each visit
- click: count of ad clicks for each visit
- (Optional column) cart_products: a comma separated text value with products added to the cart sorted by the order they were added to the cart (hint: use the sequence_number)

- This query summarizes each visit into one row showing user info, activity counts (views, cart adds, purchases, etc.), and maps it to a campaign if the visit start time falls within the campaign duration. It also lists added products in cart order.

-

visit_id	user_id	visit_start_time	page_views	cart_adds	purchase	Impression	click	cart_product	cart_products_names	campaign_name
001597	155	2020-02-17 00:21:45	11	6	1	1	1	1,4,5,7,8,9	Salmon,Russian Caviar,Black Truffle,L...	Half Off - Treat Yc
002809	243	2020-03-13 17:49:55	4	0	0	0	0	NULL	NULL	Half Off - Treat Yc
0048b2	78	2020-02-10 02:59:51	6	4	0	0	0	2,4,6,7	Kingfish,Russian Caviar,Abalone,Lobs...	Half Off - Treat Yc
004aaf	228	2020-03-18 13:23:08	7	2	1	0	0	3,7	Tuna,Lobster	Half Off - Treat Yc
006a61	420	2020-01-25 20:54:15	10	5	1	1	1	3,4,5,6,8	Tuna,Russian Caviar,Black Truffle,Ab...	25% Off - Living T
006e8c	252	2020-02-21 03:14:45	1	0	0	0	0	NULL	NULL	Half Off - Treat Yc
006f7f	20	2020-02-23 01:36:35	6	1	1	1	1	3	Tuna	Half Off - Treat Yc
007330	436	2020-01-07 22:30:36	12	8	1	1	1	1,2,3,4,5,6,...	Salmon,Kingfish,Tuna,Russian Caviar...	BOGOF - Fishing F
009e0e	161	2020-02-20 06:17:51	8	5	0	0	0	2,3,5,6,7	Kingfish,Tuna,Black Truffle,Abalone,L...	Half Off - Treat Yc
00b161	50	2020-03-15 00:24:29	12	6	1	1	1	2,3,6,7,8,9	Kingfish,Tuna,Abalone,Lobster,Crab,...	Half Off - Treat Yc

-

- [Code](#)

```
with cte as (
    select
    visit_id,
    user_id,
    min(event_time) as `visit_start_time`,
    count( distinct page_id) as `page_views`,
    sum(event_name = 'Add to Cart') as `cart_adds`,
    max(case when (event_name = 'Purchase') then 1 else 0 end)
```



```
as `purchase`,
    sum(case when (event_name = 'Ad Impression') then 1 else 0
end) as `Impression`,
    sum(event_name = 'Ad Click') as `click`,
    group_concat(
        case when (event_name = 'Add to Cart') then
product_id end
        order by sequence_number
        ) as `cart_product`,
    group_concat(
        case when (event_name = 'Add to Cart') then
page_name end
        order by sequence_number
        ) as `cart_products_names`
    from temp
    group by visit_id, user_id
)
select
    cte.*,
    cd.campaign_name
from cte , campaign_identifier cd
where
    cte.visit_start_time >= cd.start_date and
    cte.visit_start_time <= cd.end_date;
```

1. Identifying users who have received impressions during each campaign period and comparing each metric with other users who did not have an impression event

campaign_name	Impression	total_vis	avg_page_views_per_visit	avg_cart_adds_per_visit	avg_purchases
25% Off - Living The Lux Life	Not Received	300	5.5000	1.5367	2.6000
25% Off - Living The Lux Life	Received	104	9.4808	5.0962	5.3563
BOGOF - Fishing For Compl...	Not Received	195	5.3231	1.4205	2.5000
BOGOF - Fishing For Compl...	Received	65	9.6154	5.3538	5.5818
Half Off - Treat Your Shelf(sh)	Not Received	1810	5.3392	1.4912	2.4629
Half Off - Treat Your Shelf(sh)	Received	578	9.3651	5.0052	5.1481

-
- [Code](#)

```
select
campaign_name,
case when impression then 'Received' else 'Not Received' end as
Impression,
count(visit_id) as `total_visits`,
sum(page_views)/count( visit_id) as `avg_page_views_per_visit`,
sum(cart_adds)/count( visit_id) as `avg_cart_adds_per_visit`,
(sum(case when purchase=1 then cart_adds else 0 end )/sum(purchase))
as `avg_purchases`
from clique_bait_reporting.campaigns_analysis
group by campaign_name, impression
order by campaign_name;
```

2. Does clicking on an impression lead to higher purchase rates?

	clicked	purchase_rate_percentage
►	Yes	89.6494
	No	66.2162

-
- clicking on an impression lead to higher purchase rates : yes:
- [Code](#)

```
select
case when click then 'Yes' else 'No' end as `clicked`,
```

```

    (sum(purchase)/count(visit_id))*100 as
`purchase_rate_percentage`
from clique_bait_reporting.campaigns_analysis
where impression= 1
group by click;

```

3. What is the uplift in purchase rate when comparing users who click on a campaign impression versus users who do not receive an impression? What if we compare them with users who just an impression but do not click?

- uplift in purchase rate when comparing users who click on a campaign impression versus users who do not receive an impression

Result Grid				
		Filter Rows:		
		Export:		
		Wrap Cell Content:		
	Description	Impression	clicked	purchase_rate
▶	users clicked on impression	Receved	clicked	89.6494
	users not recived impression	Not Receved	not dicked	37.9176

	purchase_rate_uplift
▶	51.7318

- [Code](#)

```

select
`users clicked on impression` as `Description`,
case when impression then 'Receved' else 'Not Receved' end as
Impression,
case when click then 'clicked' else 'not clicked' end as clicked,
(sum(purchase)/count(visit_id))*100 as `purchase_rate`
from clique_bait_reporting.campaigns_analysis
where impression =1 and click = 1
union all

```

```
select
  'users not recived impression' as `Description`,
  case when impression then 'Recevier' else 'Not Recevier' end as
  Impression,
  case when click then 'clicked' else 'not clicked' end as clicked,
  (sum(purchase)/count(visit_id))*100 as `purchase_rate`
from clique_bait_reporting.campaigns_analysis
where impression= 0 and click =0;
```

- uplift in purchase rate when comparing users who click on a campaign impression versus users who just an impression but do not click

	Description	Impression	clicked	purchase_rate
▶	users clicked on impression	Recevier	clicked	89.6494
	users recived impression but not clicked	Recevier	not clicked	66.2162

- [Code](#)

```
select
  'users clicked on impression' as `Description`,
  case when impression then 'Recevier' else 'Not Recevier' end
  as Impression,
  case when click then 'clicked' else 'not clicked' end as
  clicked,
  (sum(purchase)/count(visit_id))*100 as `purchase_rate`
from clique_bait_reporting.campaigns_analysis
where impression =1 and click = 1
union all
select
  'users recived impression but not clicked' as `Description`,
  case when impression then 'Recevier' else 'Not Recevier' end
  as Impression,
  case when click then 'clicked' else 'not clicked' end as
  clicked,
```

```
(sum(purchase)/count(visit_id))*100 as `purchase_rate`
from clique_bait_reporting.campaigns_analysis
where impression= 1 and click =0;
```

4. What metrics can you use to quantify the success or failure of each campaign compared to each other?

	campaign_name	total_users	total_visits	avg_page_views_per_visit	avg_purchases_per_visit	purchase_rate
▶	25% Off - Living The Lux Life	160	404	6.5248	2.4530	50.0000
	BOGOF - Fishing For Complim...	103	260	6.3962	2.4038	48.8462
	Half Off - Treat Your Shellf(ish)	449	2388	6.3137	2.3417	49.4137

-
- [Code](#)

```
select
  campaign_name,
  count( distinct user_id) as `total_users`,
  count( distinct visit_id) as `total_visits`,
  sum(page_views)/count( visit_id) as `avg_page_views_per_visit`,
  (sum(cart_adds)/count(visit_id)) as `avg_purchases_per_visit`,
  (sum(purchase)/count(visit_id))*100 as `purchase_rate`
from clique_bait_reporting.campaigns_analysis
group by campaign_name;
```

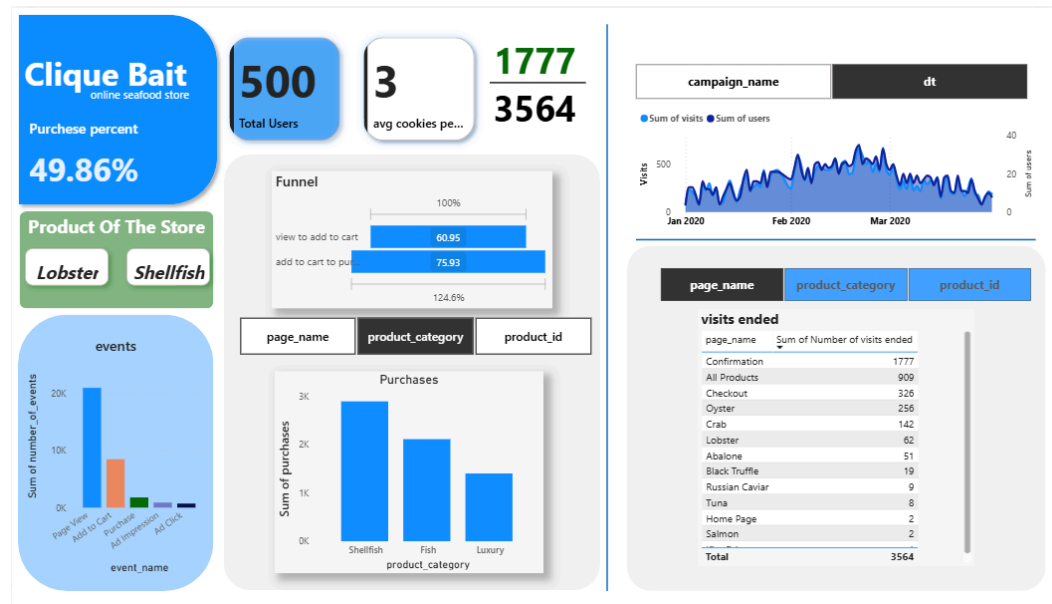
- All three campaigns show similar performance in terms of average page views per visit, average purchases per visit, and purchase rate. However, when considering total users and total visits, the "**Half Off - Treat Your Shellf(ish)**" campaign outperforms the others in absolute numbers.

5. Insights

- Total visits are **3536**
- Visits that purchased are **1777**
 - **Percentage : 49.86%**
- Total visits that has no purchase event are **1759**
 - **256 (7.2%)** of total visits ended at **Oyster page**
- On average a user visited at least 3 times

- **Half Off - Treat Your Shellfish(ish)** campaign has most visits
- On average clicking the campaign impression increases purchase rate of campaign products by **51.73%** compared to users not received Impression
- Clicking campaign impression has higher purchase rate of **89.64%**

5. Report



5. Github Repo

- [github repo](#)